

VirtuBox Infotech Private Limited

Executive Data Analyst Assessment Report

Commercial Sales & Customer Retention Analysis — Online Retail II Dataset

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LIVE ARTIFACT INDEX

- Master Google Drive Submission Folder — <https://drive.google.com/drive/folders/1TgJOIRC5My2eIWNXFeQc6UDIT6DZN9Ev?usp=sharing>
- Master Google Sheet — Data, Processed Data, Q1–Q10 — <https://docs.google.com/spreadsheets/d/1QB6U6ub7U75DF61H0o2GYxpZozNVnB7EVqBU7717MM0/edit?usp=sharing>
- Interactive Looker Studio Executive Dashboard — <https://datastudio.google.com/s/gsBLO8zDDeo>

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Executive Summary

This report presents a complete end-to-end commercial analytics assessment built on the Online Retail II dataset — 525,461 raw transaction rows spanning December 2009 to December 2011 across a UK-based non-store online retailer's international customer base. The analysis was conducted in Python (Pandas/NumPy), cleaned to 504,731 verified commercial transaction records, and delivered through a live interactive dashboard, a structured Google Sheet workbook, and a 7-slide executive presentation.

The central finding is a structural revenue concentration risk: over 85% of commercial revenue is tied to the UK domestic market, while a small set of international European accounts — particularly in the Netherlands and EIRE — generate substantially higher average order values, consistent with bulk B2B distributor behavior rather than individual consumer purchasing. Combined with sharp Q4 seasonal demand surges and a Pareto-shaped customer base (top 20% of accounts driving ~80% of revenue), this points to three priorities for management: formalizing a tiered B2B loyalty program, actively expanding European wholesale acquisition, and pre-building supply chain buffers ahead of the Q4 holiday peak.

Metric	Finding
Raw Records Processed	525,461 rows
Verified Commercial Records (Post-Cleaning)	504,731 rows
Primary Market Concentration	85%+ of revenue from UK
High-AOV International Markets	EIRE, Netherlands, Germany (up to 2.5x AOV)
Peak Seasonal Surge (Q4)	>60% order volume increase (Oct–Dec)
Customer Revenue Concentration	Top 20% of accounts ≈ 80% of revenue
Projected Impact of Recommendations	+10–15% retention · +20% international revenue

i How to read this report

Sections Q1–Q10 mirror the assessment questionnaire in full, each closing with management-facing implications. All source artifacts — the raw analysis notebook, processed dataset, and interactive dashboard — are linked throughout and consolidated in the index below for direct verification.

Live Artifact & Repository Index

The following live, hyperlinked resources contain the full underlying work product referenced throughout this report — the analysis notebook and processed dataset, the master data workbook, and the interactive executive dashboard.

#	Resource	Contents	Link
1	Master Google Drive Submission Folder	All raw deliverables: notebook, datasets, presentation, and supporting files.	Open 
2	Master Google Sheet	Raw data, processed data, and worked answers for Q1–Q7, Q8, and Q10.	Open 
3	Interactive Looker Studio Dashboard	Executive scorecard with live filters for country and date range.	Open 

QUESTION 1 | Dataset Identification & Context

Dataset Overview

Field	Detail
Dataset Name	Online Retail II (UCI Machine Learning Repository)
Source	archive.ics.uci.edu/dataset/502/online+retail+ii
Dimensions	525,461 rows × 8 columns
Columns	Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Coverage Period	December 2009 – December 2011
Business Context	Transaction logs of a UK-based non-store online retailer operating across 40+ countries

Selection Rationale

- Exceeds the 50,000-row minimum volume requirement by more than 10x, supporting statistically robust analysis.
- Contains complex, realistic data-quality issues — missing Customer IDs, order cancellations, and multi-country transactions — that mirror genuine enterprise analytics conditions.
- Enables identification of international revenue potential, exposure of revenue concentration risk, and evaluation of customer churn dynamics.

\$ Strategic Opportunities Framed by the Dataset

The breadth of country-level and time-series data allows this assessment to move beyond descriptive reporting into strategic territory: geographic diversification, seasonal operations planning, and customer retention economics.

QUESTION 2 | Business Problem, Research Questions & Hypotheses

Problem Statement

VirtuBox's underlying retail dataset reveals excessive revenue concentration in the UK market, pronounced seasonal demand volatility around Q4, and an absence of structured customer retention programs — three conditions that jointly elevate commercial risk and constrain growth.

Research Questions

ID	Research Question
RQ1	What proportion of overall revenue comes from the UK versus international markets?
RQ2	Which products and customer segments account for the top 80% of sales volume?
RQ3	How severe are Q4 seasonal demand surges on supply chain operations?

Hypotheses Tested

ID	Hypothesis
H1	International European transactions produce a significantly higher Average Order Value (AOV) than domestic UK transactions.
H2	Transaction volume surges heavily during Q4 due to pre-holiday seasonal demand.

QUESTION 3 | Data Processing & Cleaning Methodology

Approach: Python (Pandas, NumPy) via Jupyter Notebook

All cleaning and transformation logic was executed in [VirtuBox Data Analyst Assessment.ipynb](#), reducing the raw dataset from 525,461 rows to 504,731 verified commercial transaction records, published as processed_data.csv within the [master Google Sheet](#).

Stage	Row Count
Raw Dataset Volume	525,461 rows
Cleaned / Verified Volume	504,731 rows
Rows Removed in Cleaning	20,730 rows (≈ 3.9%)

Three Major Cleaning Decisions

Change	Why Necessary	Consequence if Omitted
Cancellation Filtering	Removed invoice records prefixed with 'C' (negative quantities).	Would corrupt gross volume by netting negative returns against forward sales without transaction timestamp reconciliation.
Missing Customer ID Handling	Isolated null Customer IDs to preserve statistical integrity for customer-level metrics.	Would either discard valid transaction volume or silently distort customer-level aggregates if left unhandled.
Feature Engineering	Computed line-item commercial revenue as (Quantity × Price).	Would leave no monetary basis for revenue modeling, AOV calculation, or downstream financial analysis.

QUESTION 4 | Exploratory Analysis & 5 Core Business Insights

Insight	Supporting Evidence	Management Relevance	Potential Impact	Recommendation
1. Geographic Concentration	85%+ of total commercial revenue tied to the UK market.	Signals single-market dependency risk to leadership.	High exposure to UK-specific demand or regulatory shocks.	Diversify revenue base through targeted international growth.
2. High-AOV International Markets	EIRE, Netherlands, and Germany show up to 2.5x higher basket values despite fewer orders.	Reveals an underexploited high-value customer segment.	Untapped margin and revenue upside from wholesale-style buyers.	Prioritize B2B-style acquisition in these three markets.
3. Q4 Demand Seasonality	Order volume surges >60% between October and December annually.	Directly affects inventory, staffing, and fulfillment planning.	Risk of stockouts or fulfillment delays without preparation.	Build 4–6 week inventory buffers ahead of Q4.
4. Pareto Customer Distribution	Top 20% of customer accounts generate ≈80% of aggregate revenue.	Retention of this cohort is disproportionately critical.	Losing a small number of accounts materially impacts revenue.	Launch a tiered loyalty/account-management program.
5. Catalog Skew	Top 10% of catalog items generate the majority of repeat purchase transactions.	Clarifies which SKUs merit merchandising priority.	Inventory and marketing spend may be misallocated across the long tail.	Concentrate promotional and inventory focus on top-decile SKUs.

QUESTION 5 | Surprising & Non-Intuitive Finding

Dimension	Detail
Initial Expectation	Domestic UK buyers would produce the highest Average Order Value (AOV) due to low local shipping costs and convenience.
Actual Data Finding	International European accounts — notably Netherlands and EIRE — demonstrated substantially higher basket size and AOV.
Root Cause	Overseas buyers act as B2B bulk distributors rather than individual B2C consumers.
Strategic Implication	Capital allocation should prioritize European B2B wholesale partner acquisition rather than domestic retail marketing.

\$ Why this matters

This finding directly overturns the intuitive assumption behind VirtuBox's current market approach: growth investment aimed at deepening domestic UK penetration may be less efficient per unit of spend than redirecting resources toward a small number of high-value European wholesale relationships.

QUESTION 6 | Data Quality Limitations & Analytical Risks

Identified Data Quality Issues

- Missing Customer IDs in guest checkout logs, limiting completeness of customer-level analysis.
- Absence of unit Cost of Goods Sold (COGS), preventing calculation of net profit margins.
- Lack of transit/delivery timestamps, limiting fulfillment and logistics-cycle analysis.

Explicit Analytical Limitations

- Gross revenue analysis does not reflect net operating profit margins.
- Return cycle times cannot be accurately modeled without return completion timestamps.

! Risk Disclaimer

Profitability across specific SKUs cannot safely be inferred without product manufacturing and shipping cost structures. All revenue-based conclusions in this report should be read as top-line commercial indicators, not margin or profitability guarantees.

QUESTION 7 | Actionable Strategic Recommendations for Management

#	Initiative	Description	Owner
1	Tiered B2B Loyalty Engine	Establish structured credit terms and dedicated account support for the top 20% of high-value accounts.	Commercial Sales
2	European Expansion	Target direct digital acquisition and wholesale distribution in Germany, EIRE, and the Netherlands.	Marketing & Growth
3	Pre-Q4 Supply Chain Buffering	Implement 4-to-6-week inventory buffers ahead of annual Q4 holiday transaction spikes.	Supply Chain & Operations

Target Metrics for Tracking Success

- **Customer Lifetime Value (CLV)** — tracks the long-run revenue impact of retention efforts.
- **Repeat Order Frequency** — measures traction of loyalty and account-management initiatives.
- **Net Churn Rate** — monitors whether revenue concentration risk from Q4 is being mitigated.

QUESTION 8 | Interactive Looker Studio Executive Dashboard

Live Dashboard: <https://datastudio.google.com/s/gsBLO8zDDeo>

Visual Structure

- Executive Scorecard — Total Revenue, Orders, and Unique Customers at a glance.
- Monthly Sales Time Series — visualizes seasonal trend and Q4 surge pattern.
- Top 10 Products by Revenue — ranks catalog performance.
- Geographic Concentration Donut Chart — visualizes UK vs. international revenue split.
- Country and Date Range interactive filters — allow ad hoc executive exploration.

i Purpose

Built for non-technical leadership to evaluate revenue velocity and geographic performance at a glance, without requiring direct engagement with the underlying dataset or notebook.

QUESTION 9 | Executive Management Presentation Summary

Deck: Commercial Sales & Customer Retention Analysis.pdf (7 slides)

Slide	Content
1	Executive Title & Scope
2	Business Problem & Geographic Concentration Risk
3	500k+ Record Data Pipeline & Python Architecture
4	Key Insights & High-AOV European Opportunities
5	Strategic Pillars & Action Items
6	Projected ROI (+10–15% Retention, +20% International Revenue)
7	Analytical Limitations & RFM Segmentation Roadmap

✓ **Projected ROI Highlight**

The presentation frames the commercial upside of the three recommended initiatives as a +10–15% improvement in customer retention and a +20% increase in international revenue, positioning Slide 6 as the pivotal leadership decision point.

QUESTION 10 | Responsible AI Usage Documentation

AI Systems Used

System	Functional Application
ChatGPT	Data cleaning logic and exploratory query development.
Gemini	Presentation layout generation and documentation support.
Manus AI	Documentation and cross-referencing support.

Validation Example

AI-generated aggregation syntax was cross-checked against raw Pandas groupings to prevent double-counting of refunded invoice rows — ensuring that AI-assisted code accelerated development without compromising the accuracy of reported revenue figures.

i Responsible AI Practice

All AI-assisted outputs (cleaning logic, queries, layout, and documentation) were independently verified against raw data operations before inclusion in this report, consistent with responsible-use expectations for AI-assisted analytical work.

Closing Statement & Candidate Declaration

This report consolidates the full VirtuBox Data Analyst assessment — dataset selection and rationale, cleaning methodology, exploratory findings, a counter-intuitive strategic insight, documented data limitations, actionable recommendations, the live executive dashboard, the management presentation summary, and responsible AI usage disclosure — into a single, verifiable submission.

I declare that the analysis, cleaning methodology, and conclusions presented in this report are my own work, developed with the documented assistance of AI tools as disclosed in Question 10, and validated against the underlying dataset prior to submission.

Field	Value
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